# README

## Post-bop modal-tonal AI jazz improvisation

### Program(me) Notes

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WOLFSON
COLLEGE

# Now’s the time

- Back in the late 1900s I was part of a jazz workshop helping people learn to play jazz.
- One of the exercises was called "slick licks" – we would sit in a circle and “trade licks” on our instruments.
- A big barrier for jazz learners can be finding people to play with.
- Q. Can I build a software system to enable people to practice their slick licks? And specifically… can a bass player use this?!
- It should listen and respond, sometimes lead, be realistic enough to be useful.
- NB This interactive “robot” is designed for practice purposes, and it's not intended to replace a human player. Human interaction was key to slick licks!

# Program(me) Note

*README** is a live conversation between a human bassist and an artificial musician that listens, responds, and remembers. Wolfson is a computer system trained on hundreds of jazz saxophone solos; it hears each bass phrase, analyses its character – the notes, the rhythm, the energy – and improvises a reply in real time. Over five minutes the music traces a journey through jazz history: an exploratory, chromatic opening gives way to the modal sound of Miles Davis and Herbie Hancock, builds to a hard-driving peak, and settles into a meditative close. No two performances are identical — the music that emerges depends entirely on what the bassist chooses to play.

* The title works on three levels: the *README* file is the first thing you read when you encounter a piece of software; the software reads me, the bassist, responding to whatever I play; and it is a sideways glance at *Recorda Me* ("remember me") a jazz standard, and a reminder that all improvisation is, in some sense, an act of memory.

# All the things I have been* thinking about

Phrase analysis
Question and answer
Leadership & role
Opening echo
Contour steering
Chord conditioning
Harmonic modes
Scale pitch bias
Pitch range
Swing / triplet feel
Live tempo tracking
Dynamics
Generation temperature

Phrase-shape dynamics
Articulation
Bass pitch-class tracking
Scale mode detection
Energy arc
Motivic development
Rhythmic displacement
Lyrical motif re-use
Voice leading
Modal leap bonus
Stochastic thinning
Phrase breathing
Singable duration bias

Rhythmic density
Rhythmic complementarity
Register contrast
Phrase length control
Beat-matching / trading bars
Repetition control
Sax riff / insistence
Proactive mode
Live phrase peeking
Beat-synchronised entry
Turn-taking
Loop mode
Arc timing

** Good name for a song?*

# Fly me to the moon

**The five giant steps in the performance arc…**

| | |
|---|---|
| **Sparse** | Exploratory, establish motifs slowly |
| **Building** | Groove warming up, modal |
| **Peak** | Fast, active, max rhythmic intensity |
| **Recapitulation** | Lyrical return, motifs recalled |
| **Resolution** | Slowest, long tones, fading out |

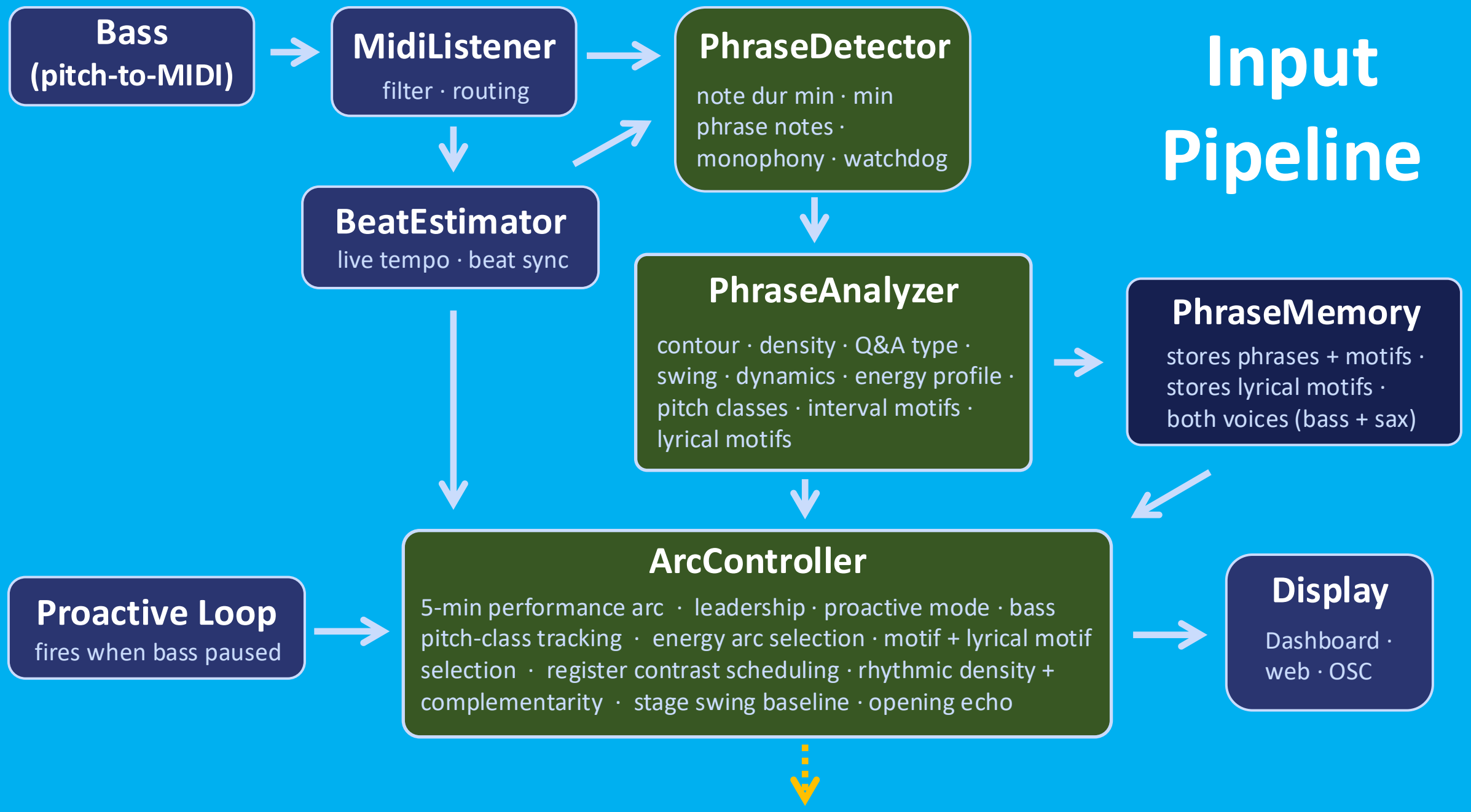
# Sentimental journeys

This is inspired by:

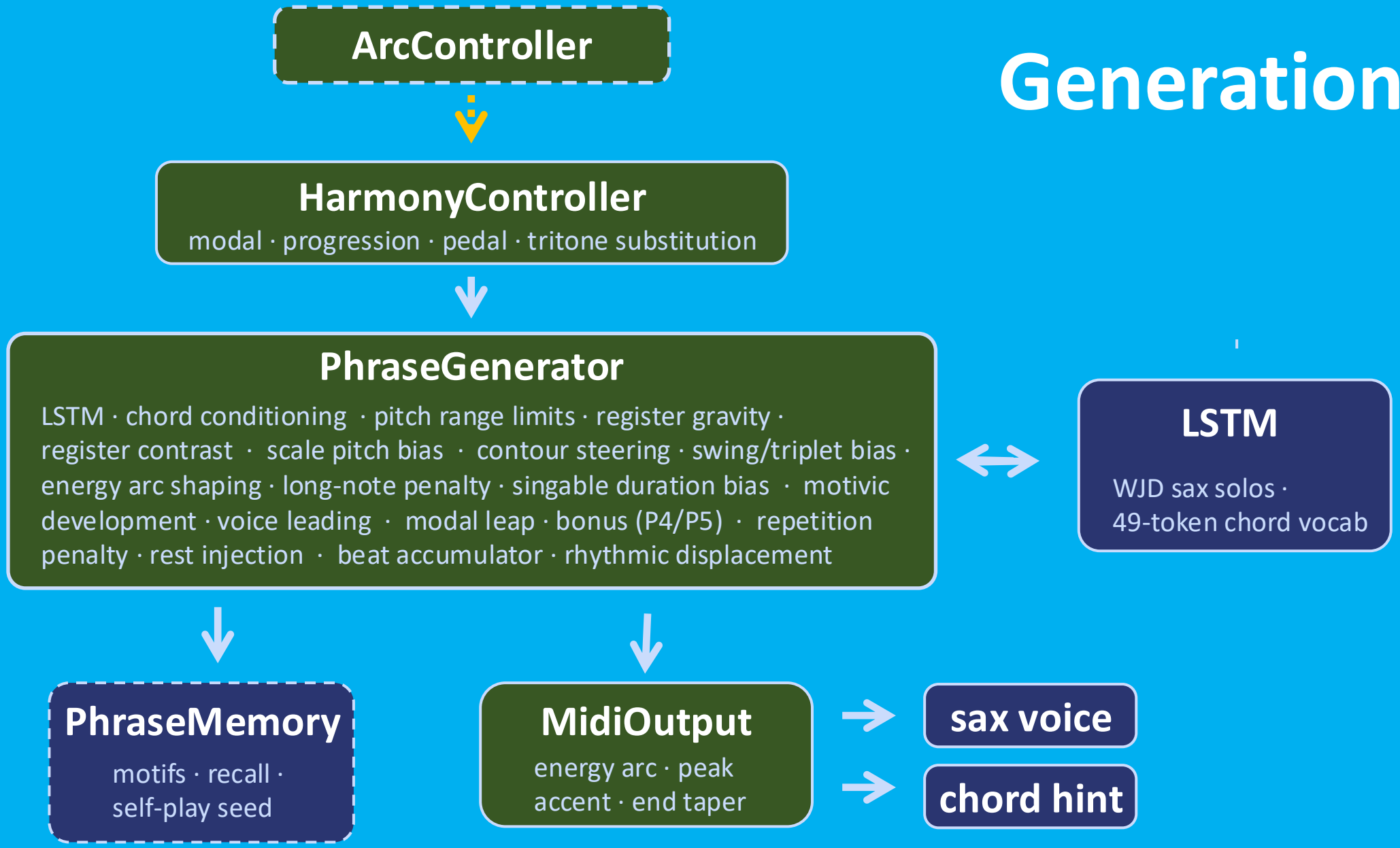
- *Post-bop modal-tonal hybrids* like:
  - **Milestones** (Miles Davis), **Little Sunflower** (Freddie Hubbard), **Maiden Voyage** (Herbie Hancock), **Footprints** (Wayne Shorter), and the atmosphere of **In a Silent Way** (Miles Davis). These songs are my favourite things*.
- The arc also traces a journey through jazz history :
  - Sparse opening resembles the chromatic freedom of free jazz,
  - Modal building stage reflects the Miles/Coltrane modal period,
  - Peak combines hard bop drive with modal intensity,
  - Pedal resolution has the meditative quality of late Coltrane e.g. A Love Supreme.

* *John Coltrane*

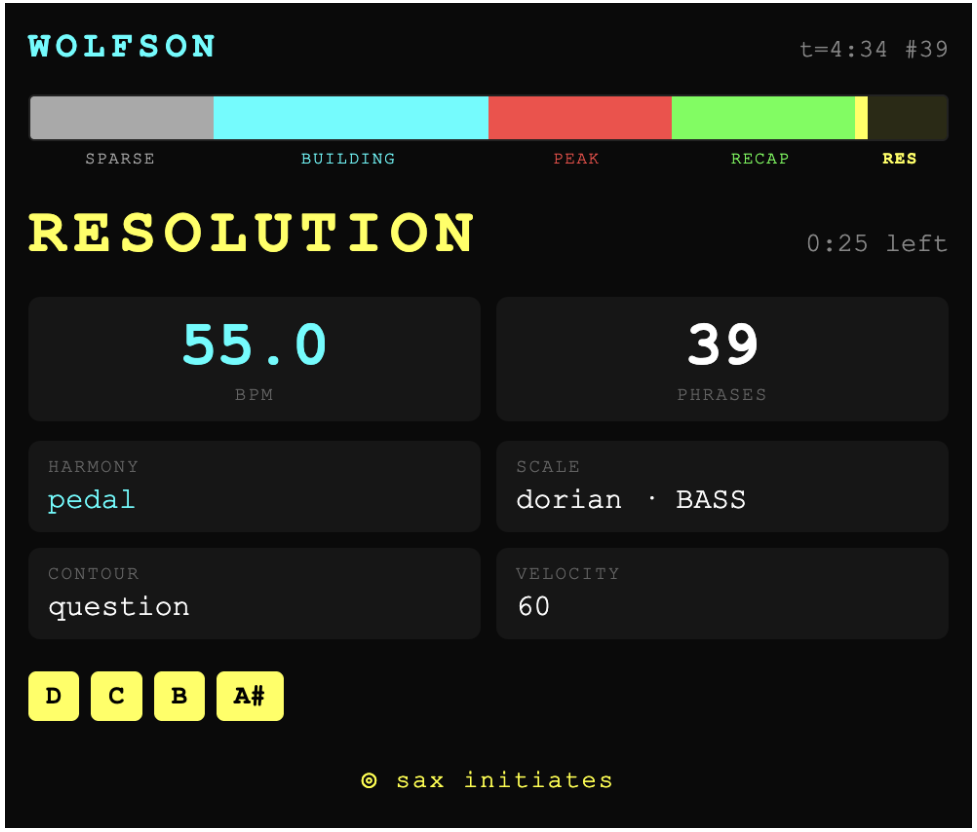
# Input Pipeline



# Generation


ArcController
HarmonyController
modal · progression · pedal · tritone substitution
PhraseGenerator
LSTM · chord conditioning · pitch range limits · register gravity · register contrast · scale pitch bias · contour steering · swing/triplet bias · energy arc shaping · long-note penalty · singable duration bias · motivic development · voice leading · modal leap · bonus (P4/P5) · repetition penalty · rest injection · beat accumulator · rhythmic displacement
LSTM
WJD sax solos · 49-token chord vocab
PhraseMemory
motifs · recall · self-play seed
MidiOutput
energy arc · peak accent · end taper
sax voice
chord hint

# Soul A-eyes



| — | BASS (you) | SAX |
|---|---|---|
| Phrases | — | 44 |
| Notes | — | 322 |
| Note length | — | 0.51b |
| Short notes | — | 51% |
| Pitch range | — | F3 – D6 |
| Dynamics | — | 80 – 80 |

Watch live during the performance on

**wolfson.numbersintonotes.net**

- Repo is on github.com/davidderoure/wolfson


- Acknowledgement:
  - Ray d'Inverno (for jazz, Lisp, AI, slick licks)
  - Mark d'Inverno (for jazz, AI, creativity, slick licks)
  - George E Lewis (for Voyager and Forager)
  - Robert Laidlow (for Alter-ing Ada Lovelace, and sax)
  - Hongshuo Fan (for Music Gesture Recognition)
  - Bofan Ma (for always enabling musicality)
  - Maria Kallionpää (for Climb!)
  - Ada Lovelace & Emily Howard (for Numbers into Notes)
  - Wolfson Digital Research Cluster, DiSc & TORCH (for AIMC 2024)


- *Grand Challenges for Computing Research* – Tony Hoare

# Radar of the lost archive

- "The Grand Challenges Exercise of the UK Computing Research Committee"
- A workshop held in Edinburgh in November 2002, considered 109 submissions.
- Archival research suggests my submission was "The creative electronic musician". Or maybe "Interactive Electronic Musician".
- Sir Tony Hoare, Wolfson Emeritus Fellow, was Chair of the programme committee.

National e-Science Centre

Workshop on Grand Challenges for Computing Research

*Sponsored by the UK Computing Research Committee, with support from EPSRC and NeSC*

UK COMPUTING RESEARCH COMMITTEE

**Panel C Submissions**

5. Deep Adaption
6. Virtual Realms
7. If we could see fit
8. The creative electronic musician
9. Getting Androids to learn from testimony
10. Rendering with intent
11. Deep Thought

# Someday my footprints will come but we might not be quite there yet:

- It uses MIDI: no pitch inflection, vibrato, or continuous dynamics within notes – every note is a rectangle.
- Time feel is mechanical within phrases: no microtiming, no laying back or pushing the beat.
- Conversation is polite by design – complementary, never confrontational, doesn't interrupt, crowd the register, or refuse to listen..
- It's pretty much in 4/4, no waltzes or take 5 – the tempo estimator assumes equal beats.
- No personal voice – an amalgam across an era rather than any one player's sound.
- The arc is clock-driven, not music-driven – deliberately for this performance (but see *Climb!*)
- Motivic development might feel like repetition – because it is.
- Trained on solos not melodies – no quoting, no playing off the tune.
- Trained on one era and school – the bebop/hard-bop vocabulary of the Weimar Jazz Database; no awareness of what came before or after.
- Inside/outside distinction and narrative arc within a phrase could be designed more explicitly – the model has learned some of this from transcriptions.
- It only listens to one instrument (but see *Forager)*.
- Audiences may find fixed, familiar harmony more immediately satisfying – responsiveness to the bassist means harmony is earned rather than given.

# Wolfson — Performance Plan

Bass improvisation cues by arc stage — 1 May 2026, Wolfson College Oxford

**Before playing (spoken):** *"I'm going to have a musical conversation with an AI. I play the bass — it plays the saxophone. Watch how it listens and responds to what I play."*

| Sparse<br>0:00–1:00 | Building<br>1:00–2:30 | Peak<br>2:30–3:30 | Recapitulation<br>3:30–4:30 | Resolution<br>4:30–5:00 |
|---|---|---|---|---|
| • **Phrases echo back literally** — same intervals and rhythm, transposed to sax register. Play short, distinctive phrases.<br>• **Leave 2+ beats of rest** after each phrase — system needs ~2 s silence to close the phrase and respond.<br>• **Play quietly** — sax mirrors your dynamic. A quiet opening makes later contrast dramatic.<br>• **Plant a 2–3 note motif** — it will be recalled in recapitulation.<br>• *Musical material: 3–5 note fragment — So What figure (D–F–G–A); A Love Supreme motif (A–B♭–C–E); Fly Me to the Moon opening (E–G–A–B–C — Sinatra, cross-audience). Transpose freely; audiences recognise shape, not key. Echo lands on sax immediately. Keep it melodic; chromatic lines are harder to follow.* | • **Play D Dorian** (D E F G A B C) — 4+ distinct pitch classes shifts the sax harmonic language. Switch away briefly to hear it change.<br>• *Modal groove **continuously for 8–10 s** → sax enters trading mode. Options — So What vamp (D Dorian); Cantaloupe Island F minor vamp (skip D♭/D♮ changes — highest recognition via US3 sample); Footprints in 4/4 (C minor Dorian); Watermelon Man funky hook (widely sampled, suits trading mode). Pitch-class tracking shifts the sax to whichever tonality you establish.*<br>• *Melodic lick → echo: Autumn Leaves descending (G–F–E♭–D–C, stepwise, echoes cleanly — most recognised jazz melody); Summertime opening 4 notes (minor pentatonic — widest cross-audience recognition). Play short, rest 2 beats, hear the reply.*<br>• **Repeat a riff 3+ times** — sax enters development mode: stronger motivic use, directed contour. It starts to *argue with* the riff.<br>• *Late building (~2:00): introduce a bluesy bass lick (D blues scale: D–F–G–A♭–A–C) to seed the peak blues character.*<br>• **Let the sax repeat** (`--sax-riff-prob`) — sax insists on its phrase; you develop underneath. After two replays it varies itself. | • *Blues bassline (loud, driving): sax is in blues mode — harmonic match. Options — 12-bar figure (universally legible); Moanin' (Timmons/Blakey) 4-note gospel riff (physical, accessible, call-and-response obvious); All Blues (Miles Davis) 3-note figure (same Kind of Blue world as So What).*<br>• **Play loudly and densely, then sparsely** — busy bass invites sparser sax; a terse phrase invites a rush of notes.<br>• **Rising phrase then falling** — contour steering. Sax responds to a rising line with a descending resolution and vice versa.<br>• **Switch between Q & A and continuous playing** — short phrases with 2-beat rests give call-and-response; sustained riff triggers trading mode. The contrast is most audible here. | • **Play low** — register contrast pushes sax into upper register. Spatial separation is most audible here.<br>• **Listen** — sax quotes motifs from sparse and building stages. Pause to let the audience hear the system remembering.<br>• *Return quietly to the building groove figure — sax may quote it back (the AI has remembered). Alt.: Little Sunflower (F major/Lydian, very slow) — lyrical character suits recap perfectly and gives the sax room for long tones.* | • *One final long bass note, then silence. Put the bass down deliberately.*<br>• **Sax closes the piece unaccompanied** — the clearest demonstration of musical agency. Let it finish. |